

ICML 2023: Dropping the Gaussian-Noise Assumption in Gravitational-Wave Inference

A Montréal and Flatiron Institute team introduces Score-based Likelihood Characterization (SLIC), learning real detector noise directly from data instead of assuming its shape.

Almost all gravitational-wave data analysis rests on a convenient fiction: that detector noise is Gaussian and stationary. The assumption is what makes the likelihood tractable and cheap to evaluate, and it has underpinned most of the last decade's parameter estimation. Real instruments are less obliging. The sensitivity of LIGO, Virgo, and KAGRA drifts over time, their data are peppered with transient bursts called glitches, and narrow spectral lines wander slowly across the band.

These departures are not academic. The most famous example is the first binary neutron star merger, GW170817, which landed squarely on top of a loud glitch in a LIGO detector and required bespoke data surgery before it could be analyzed. As gravitational-wave astronomy moves into an era of large event catalogs, even small deviations from the Gaussian assumption quietly accumulate into systematic biases across thousands of signals.

This work, from the Université de Montréal, Mila, Ciela, and the Flatiron Institute, takes a different route. If real noise is hard to write down in closed form, then stop assuming it and learn it from the data instead. The authors bring a technique called Score-based Likelihood Characterization (SLIC) to gravitational-wave inference: a score-based diffusion model learns the distribution of real detector noise, which is then coupled to a differentiable physical waveform model to form an unbiased likelihood that never assumes the noise is Gaussian.

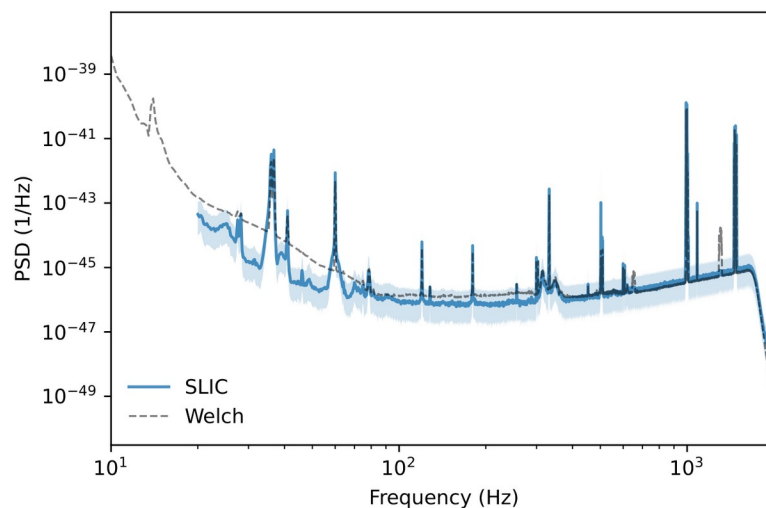


Figure 1. Power spectral density (PSD) of synthetic noise generated by SLIC (blue), compared with a Welch estimate from real LIGO-Hanford data (dashed gray). The blue curve is drawn from 1024 synthetic segments,

with the light-blue band showing the 68% interval. SLIC closely matches the real noise and, having seen more data (11 h vs 1 h), resolves the narrow spectral lines even more finely than the Welch estimate.

Paper and code: the paper is available at <https://ml4astro.github.io/icml2023/assets/70.pdf> and the code is open-sourced at <https://github.com/RonanLegin/slicgw>.

The problem: the likelihood needs a noise model, and real noise has no closed form

Extracting astrophysics from a gravitational-wave detector is a Bayesian inference problem: given a stretch of data, infer the posterior over the source parameters (black hole masses, spins, distance, and so on). That step requires a likelihood, and the shape of the likelihood is dictated entirely by the statistics of the noise. The Gaussian-stationary assumption is popular precisely because it yields a closed-form likelihood that is fast and stable to evaluate.

The catch is that the glitches, spectral lines, and subthreshold astrophysical signals in real data mostly have no ready generative model. Today they are handled segment by segment with custom fixes that are expensive and can quietly introduce bias. What the authors really want is to keep every advantage of the deterministic waveform models that standard inference relies on, while dropping the one unrealistic assumption about the detector noise.

The method: learn the score of the noise instead of assuming its form

SLIC targets a class of additive-noise inverse problems, where an observation equals a deterministic signal model plus noise: $x_O = M(\theta) + x$. Here $M(\theta)$ is the waveform determined by the source parameters θ , and x is the noise. The key observation is that if you can obtain the score of the noise distribution Q — the gradient of its log-probability, $\nabla_x \log Q(x)$ — then, combined with the Jacobian of the waveform model $\nabla_\theta M(\theta)$, you can write the gradient of the likelihood with respect to the source parameters as the product below:

$$\nabla_\theta \log p(x_O | \theta) = -\nabla_x \log Q(x) \cdot \nabla_\theta M(\theta), \text{ with } x = x_O - M(\theta).$$

The method therefore splits in two. For the noise, a score-based diffusion model is trained with denoising score matching to learn $\nabla_x \log Q(x)$ directly from samples of real LIGO noise, with no assumption that it is Gaussian. For the physics, the waveform is the differentiable IMRPhenomD model implemented in the ripple package, which makes automatic differentiation — and hence the Jacobian the equation needs — possible. Multiplying the two gives the score of the likelihood, and the authors draw posterior samples from it using a Metropolis-adjusted Langevin algorithm (MALA). The framework was originally developed for astronomical imaging with Hubble and Webb; this work ports it to gravitational waves.

Check one: does the learned noise look real?

The first thing to confirm is whether the diffusion model has actually captured the statistics of LIGO noise. The authors draw 1024 four-second synthetic noise segments, each sampled at 4096 Hz, from the trained model, compute their power spectral density, and compare it against a Welch estimate from real LIGO-Hanford data (Figure 1). The two agree closely. More

strikingly, SLIC resolves the narrow, sharp spectral lines even more finely than the Welch estimate, for a simple reason: it saw 11 hours of data during training, against 1 hour for the empirical estimate. That suggests the model is learning the spectral structure of the noise, not memorizing any single stretch of data.

Check two: inject a known signal into real noise and recover it

The sharper test is end-to-end parameter estimation. The authors inject a simulated signal modeled on GW150914, with chirp mass $M = 29 M_{\odot}$, symmetric mass ratio $\eta = 0.2495$, zero spins, and luminosity distance $d_L = 400$ Mpc, into a stretch of real LIGO noise recorded after the training window and never seen by the model, and then infer all 5 of its source parameters with SLIC. As a control, they run the same inference with a standard Gaussian likelihood built from a Welch PSD.

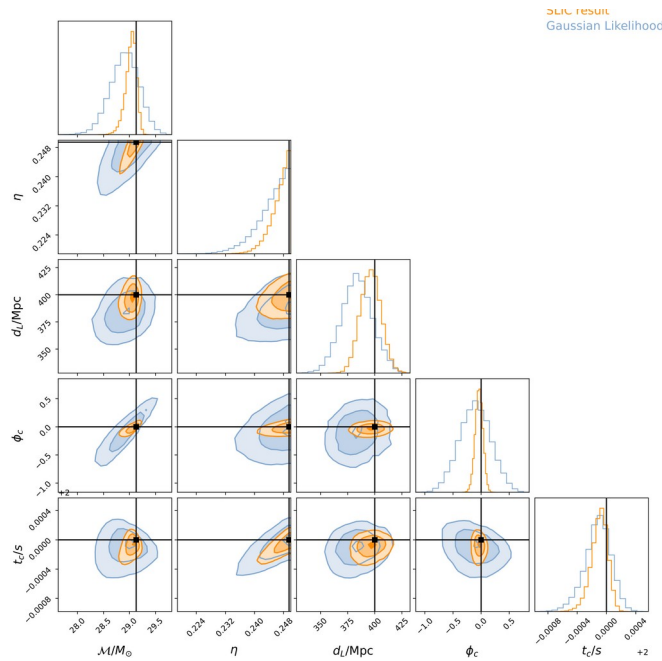


Figure 2. Posterior over the five recovered parameters of the injected signal: SLIC (orange) versus the standard Gaussian likelihood (blue). Contours mark the 10%, 39.35%, and 90% intervals, and the black lines show the true values. Both methods bracket the truth, but the SLIC posterior has a smaller variance. Parameters are the chirp mass M , symmetric mass ratio η , luminosity distance d_L , coalescence phase ϕ_c , and coalescence time t_c .

The result is Figure 2. To keep the comparison fair, the authors deliberately chose a glitch-free segment, so that even the Gaussian likelihood should recover the truth — and it does. Both methods recover all five true values with high credibility. That agreement matters: it calibrates SLIC against a trusted baseline and shows the learned likelihood introduces no bias. On top of that, the SLIC posterior comes out with a smaller variance, hinting at the gains to expect once the method is turned loose on genuinely non-Gaussian, glitch-contaminated data.

A few numbers

Table 1. Key settings and results of the SLIC gravitational-wave demonstration.

Item	Value
Training data	11 hours of real LIGO-Hanford noise around GW150914
Sampling / segmentation	4096 Hz, split into 4 s segments (the true-signal segment discarded)
Noise validation	1024 synthetic segments; PSD matches the real data
Recovered parameters	5: M , η , dL , ϕ_c , t_c
Injected signal	$M = 29 M_\odot$, $\eta = 0.2495$, $\chi_1 = \chi_2 = 0$, $dL = 400$ Mpc
Waveform model	Differentiable IMRPhenomD (ripple package)
Sampler	MALA (Metropolis-adjusted Langevin)

Why it matters, and where it stops

The core message is clean: you no longer have to assume the noise is Gaussian and stationary. By learning the real noise distribution with a diffusion model and coupling it to a differentiable waveform, SLIC builds an unbiased gravitational-wave likelihood directly from detector data, and it recovers the true parameters on real LIGO noise. Compared with segment-by-segment custom data treatment, that is a more unified and more scalable path.

It is worth being honest about the boundary. This is a proof of concept, limited to 4-second, single-detector segments, and the sampler is a plain MALA — which will bottleneck on the full compact-binary problem, where the posterior lives in roughly fifteen dimensions with nontrivial structure. The authors point to three extensions already under way: longer segments to cover neutron-star signals, multi-detector models for coherent analysis, and more powerful samplers such as flowMC. With those pieces in place, machine-learning noise models could deliver fast, scalable, and unbiased inference for the flood of events expected from next-generation detectors like Cosmic Explorer and the Einstein Telescope.